

Question description

There is a bit string consisting of n bits. Then, there are some changes that invert one given bit. Your task is to report, after each change, the length of the longest substring whose each bit is the same.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $1 \leq m \leq 2 \cdot 10^5$
- $1 \leq x_i \leq n$

Input Format

The first input line has a bit string consisting of n bits. The bits are numbered $1, 2, \dots, n$.

The next line contains an integer m : the number of changes.

The last line contains m integers $x_1, x_2, \dots, x_m$ describing the changes.

Output Format

After each change, print the length of the longest substring whose each bit is the same.

▼ Logical Test Cases

Test Case 1

INPUT (STDIN)

001011
3
3 2 5

EXPECTED OUTPUT

4 2 3

Test Case 2

INPUT (STDIN)

101010
4
1 3 2 6

EXPECTED OUTPUT

2 4 2 2

▼ Mandatory Test Cases

Test Case 1

KEYWORD

void pull(int k,int
l,int r)

▼ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

2

Test Case 2

TOKEN COUNT

550

Test Case 3

NLOC

59

```
1 #include <stdio.h>
2 #include <string.h>
3
4 #define MAXN 200005
5
6 char s[MAXN];
7
8 int max(int a, int b) { return a > b ? a : b; }
9
10 int getLongest(int n) {
11     if (n == 0) return 0;
12     int max_len = 1;
13     int cur_len = 1;
14     for (int i = 1; i < n; i++) {
15         if (s[i] == s[i - 1]) {
16             cur_len++;
17         } else {
18             if (cur_len > max_len) max_len = cur_len;
19             cur_len = 1;
20         }
21     }
22     if (cur_len > max_len) max_len = cur_len;
23     return max_len;
24 }
```

```
26void pull(int k, int l, int r) {
27
28}
29
30int main() {
31    if (scanf("%s", s) != 1) return 0;
32    int n = strlen(s);
33    int m;
34    if (scanf("%d", &m) != 1) return 0;
35
36    for (int i = 0; i < m; i++) {
```

```
37     int idx;
38     scanf("%d", &idx);
39     idx--;
40     if (s[idx] == '0') s[idx] = '1';
41     else s[idx] = '0';
42
43     printf("%d%c", getLongest(n), (i == m - 1) ? '\n' : ' ');
44 }
45 return 0;
46}
47
```

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt

001011

3

3 2 5

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Queue2.exe

4 2 3

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt

101010

4

1 3 2 6

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Queue2.exe

2 4 2 2

C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>